Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2022	ENGI3291-WE01

Title: (BEng/MEng)

Thermodynamics and Fluid Mechanics 3

Time Allowed:	2 hours
Additional Material provided:	Yes, periodic table included in exam questions
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam, submission is time-limited to 4 hours within a 24 hrs window. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the relevant 24-hour window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Q1

The axial compressor of a jet engine is being designed to have multiple repeating stages with constant axial flow velocity.

(a) Explain what is meant by a repeating stage. Why are repeating stages often used in the design of multistage axial compressors? What feature of the blade geometry needs to change gradually for the downstream stages?

[20%]

(b) Sketch the cascade view of a typical axial compressor stage and the corresponding velocity triangles. Starting from the definition of flow coefficient and reaction, show that:

$$\frac{1-2R}{\phi} = \tan\beta_1 + \tan\alpha_2,$$

where the symbols and the subscripts have the usual and same meaning as those in the handout. Hence show that if $R = 0.5$, the rotor blades are a mirror image of the stator blades.

[40%]

(c) If the exit velocity of each stage has a very small but nonzero swirl, and the reaction R is very close but not equal to 0.5, show that R can be written as:

$$R = \frac{1}{2}\left(1 - \phi\beta_2 - \phi\alpha_1\right),$$

where the symbols and the subscripts have the usual and same meaning as those in the handout. Hence sketch the possible shape of the blades in rotor and stator.

[20%]

(d) This compressor has an overall stagnation pressure ratio of 15. All stages are to have a mean blade speed of 250 m/s and a stage loading coefficient of 0.4. The total to total efficiency is estimated to be 0.85. Calculate the number of stages needed when the inlet stagnation temperature is 300K, assuming the flow is that of an ideal gas.

[20%]

continued

Q2

(a) Using the periodic table of the elements provided in Figure Q2, determine how many neutrons and protons are in the nuclei of the following atoms: Mg-25, Xe-128, Rn-224.

[10%]

(b) How does the binding energy per nucleon change in an energy releasing nuclear reaction and why? What are the two major possibilities of an exothermic nuclear reaction and why?

[10%]

(c) A reactor core consists of a very dilute homogenous mixture of natural uranium and heavy water moderator with the following properties:
Average number of neutrons per fission from ^{235}U, $\nu = 2.5$
Fission cross section for ^{235}U, $\sigma_f(^{235}\text{U}) = 580$ b
Absorption cross section for ^{235}U, $\sigma_a(^{235}\text{U}) = 700$ b
Absorption cross section for ^{238}U, $\sigma_a(^{238}\text{U}) = 2.7$ b
Absorption cross section for D_2O , $\sigma_a(D_2O) = 1.3 \times 10^{-3}$ b
Abundance (atom-percent) in natural uranium for ^{235}U, $\gamma = 0.0072$
Abundance (atom-percent) in natural uranium for ^{238}U can be taken as $1 - \gamma = 0.9928$
Taking into account that the moderator-to-fuel ratio by mass is 5.0 calculate the ratio of the number of molecules of D_2O to the number of all atoms of uranium in the mixture.

[15%]

(d) Taking into account that the number of all atoms of uranium per unit volume is 5×10^{28} atoms/m^3, calculate the total macroscopic cross-sections for absorption in the fuel and in the moderator.

[15%]

(e) Assuming the resonance escape probability, p, is 0.9 and the combined probability of fast and thermal leakage, $\chi_f \cdot \chi_t$, is 0.95 calculate the reproduction constant. Comment if neutron chain reaction is sustained.

[15%]

(f) Describe the types of steam turbine cylinders and steam conditions that are found in the steam cycles of nuclear power stations with CANDU reactors.

[20%]

(g) Summarise in 250 words or less, what role you think generating electricity from nuclear power should play in future low-carbon energy systems.

[15%]

continued

Periodic Table of the Elements

1 IA 1A	2 IIA 2A	3 IIIB 3B	4 IVB 4B	5 VB 5B	6 VIB 6B	7 VIIB 7B	8	9 VIII 8	10	11 IB 1B	12 IIB 2B	13 IIIA 3A	14 IVA 4A	15 VA 5A	16 VIA 6A	17 VIIA 7A	18 VIIIA 8A
1 H Hydrogen 1.008																	2 He Helium 4.003
3 Li Lithium 6.941	4 Be Beryllium 9.012											5 B Boron 10.811	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
11 Na Sodium 22.990	12 Mg Magnesium 24.305											13 Al Aluminum 26.982	14 Si Silicon 28.086	15 P Phosphorus 30.974	16 S Sulfur 32.066	17 Cl Chlorine 35.453	18 Ar Argon 39.948
19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.631	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798
37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium 98.907	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.906	46 Pd Palladium 106.42	47 Ag Silver 107.868	48 Cd Cadmium 112.414	49 In Indium 114.818	50 Sn Tin 118.711	51 Sb Antimony 121.760	52 Te Tellurium 127.6	53 I Iodine 126.904	54 Xe Xenon 131.294
55 Cs Cesium 132.905	56 Ba Barium 137.328	57-71	72 Hf Hafnium 178.49	73 Ta Tantalum 180.948	74 W Tungsten 183.84	75 Re Rhenium 186.207	76 Os Osmium 190.23	77 Ir Iridium 192.217	78 Pt Platinum 195.085	79 Au Gold 196.967	80 Hg Mercury 200.592	81 Tl Thallium 204.383	82 Pb Lead 207.2	83 Bi Bismuth 208.980	84 Po Polonium [208.982]	85 At Astatine 209.987	86 Rn Radon 222.018
87 Fr Francium 223.020	88 Ra Radium 226.025	89-103	104 Rf Rutherfordium [261]	105 Db Dubnium [262]	106 Sg Seaborgium [266]	107 Bh Bohrium [264]	108 Hs Hassium [269]	109 Mt Meitnerium [278]	110 Ds Darmstadtium [281]	111 Rg Roentgenium [280]	112 Cn Copernicium [285]	113 Nh Nihonium [286]	114 Fl Flerovium [289]	115 Mc Moscovium [289]	116 Lv Livermorium [293]	117 Ts Tennessine [294]	118 Og Oganesson [294]

Lanthanide Series	57 La Lanthanum 138.905	58 Ce Cerium 140.116	59 Pr Praseodymium 140.908	60 Nd Neodymium 144.243	61 Pm Promethium 144.913	62 Sm Samarium 150.36	63 Eu Europium 151.964	64 Gd Gadolinium 157.25	65 Tb Terbium 158.925	66 Dy Dysprosium 162.500	67 Ho Holmium 164.930	68 Er Erbium 167.259	69 Tm Thulium 168.934	70 Yb Ytterbium 173.055	71 Lu Lutetium 174.967
Actinide Series	89 Ac Actinium 227.028	90 Th Thorium 232.038	91 Pa Protactinium 231.036	92 U Uranium 238.029	93 Np Neptunium 237.048	94 Pu Plutonium 244.064	95 Am Americium 243.061	96 Cm Curium 247.070	97 Bk Berkelium 247.070	98 Cf Californium 251.080	99 Es Einsteinium [254]	100 Fm Fermium 257.095	101 Md Mendelevium 258.1	102 No Nobelium 259.101	103 Lr Lawrencium [262]

Figure Q2: Periodic table of the elements

continued

Q3

(a) As the free-stream Mach number tends to infinity, show that the deflection angle δ of an oblique shock wave in a perfect gas satisfies the following equation:

$$\tan\delta = \frac{\sin 2\beta_1}{\gamma + \cos 2\beta_1},$$

where the symbols have the same meaning as those in the handout. Hence calculate the maximum deflection angle δ_{max} for $\gamma = 1.4$.

[20%]

(b) A supersonic business jet is cruising at an altitude of 12 000 m above sea level at its designed cruising speed, Mach number 1.8. This aircraft is powered by a pair of engines, installed side-by-side in a centreline ventral pod. Each engine is fed by its own intake. A close-up view of the intake system is sketched in Figure Q3A, where only half of the system is shown due to symmetry. It comprises a double-wedge (AB and BC) followed by a subsonic diffuser. Figure Q3A shows two oblique shock waves which form at the Cone surface and intersect (focus) at the cowl lip E. There is no shock reflection inside the diffuser. At the design condition, the flow is subsonic everywhere downstream of the plane ED. Neglect any friction and heat transfer effects.

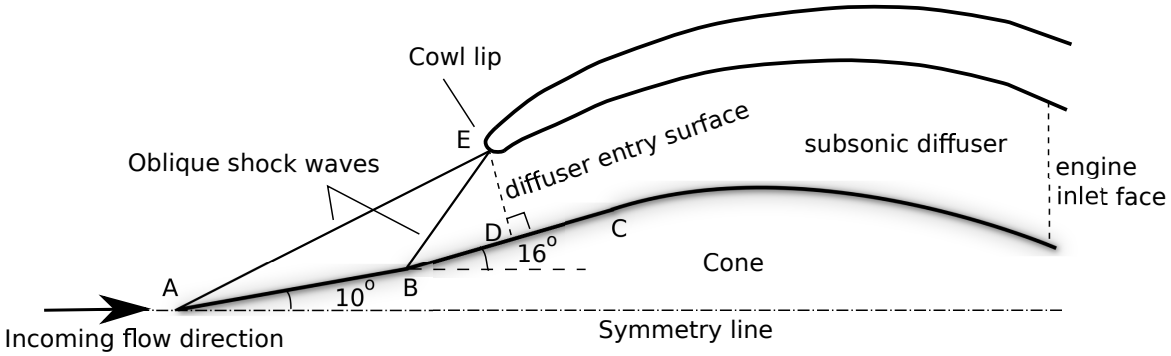


Figure Q3A: Engine at on-design condition (not to scale)

Calculate the shock wave angles with respect to the symmetry line; the Mach number immediately after ED and the stagnation pressure ratio across the intake shock system.

[50%]

(c) For the same jet in (b), calculate the absolute speed of the jet and the static and stagnation pressures measured by a Pitot tube mounted on the jet. The Pitot tube inlet section is shown in Figure Q3B; $\beta_p = 60°$.

[30%]

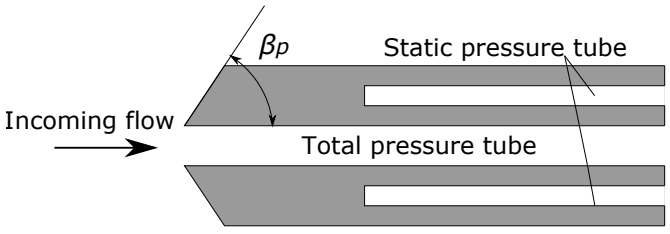


Figure Q3B: Pitot tube inlet section.

END